

Emir Burak Önder

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Professional Experience

Wi-Fi Application Software Engineer | Airties

2025/07 – 2026/07

- **Engineered C/C++ Linux-based applications** for routers and access points within **embedded Linux** environments, optimizing core **data-handling packages** to enhance device stability and overall system performance.
- **Designed and implemented standard-compliant data models (TR-181)** and integrated remote device management paths via **TR-069 (CWMP)**, ensuring seamless provisioning, telemetry, and configuration of Wi-Fi devices.
- **Engineered core components for the Device Identification engine**, utilizing **Netfilter and iptables** to build robust packet capture pipelines. Processed and analyzed the intercepted cross-protocol traffic (**DHCP, mDNS, ARP, DNS, HTTP, and SSDP**) to significantly enhance device classification accuracy and inspection throughput.
- **Built low-level networking and diagnostic utilities** using advanced **socket programming**, creating robust packet generation, network analysis, and debugging tools tailored for embedded systems.
- Modernized and refactored the core architecture of the Device Identification package using **modern C++ standards**, authoring a comprehensive unit test suite to achieve over **88% code coverage**.

Network Engineer | Labris Networks

2024/10 – 2025/07

- Developed a high-performance kernel-level firewall using **C** and **eBPF/XDP**, building **kernel-space packet processing** logic and **user-space management applications**. Designed and implemented **application-aware traffic filtering** and configurable **blacklist** enforcement based on security policies.
- Integrated **nDPI/nDPId** for **real-time application detection** and classification, enabling **policy-based filtering** and **multi-WAN** traffic steering.
- Developed backend services and **REST APIs** using **Python (Django & Flask)**, with **PostgreSQL** database integration and **Nginx**-based deployment.

Linux System Administrator | Ostim Technical University

2023/11 – 2024/06

- I have experience working on **Identity Management** using **Keycloak**, with strong skills in **Linux**, **Odoo**, and **PostgreSQL** on **Debian**.

Linux System Administrator Intern | Özgür Yazılım A.Ş.

2024/06 – 2024/09

- Managed Linux-based infrastructure (**Rocky Linux, Debian**), supporting CI/CD with Jenkins, web services with **Nginx**, monitoring with **Zabbix**, and self-hosted **GitLab** administration.

Software Engineer Intern | Tubitak Ulakbim (Pardus)

2023/05 – 2023/09

- Built a Debian repository management tool ☑ using **Python/Django** and **APTLY**, implementing **REST APIs** and **real-time process streaming**.
- Ported the **Debian Package Tracker** to **Pardus**, contributing to the adaptation and maintenance of an package tracking system.

Teaching & Specialized Training

Co-Instructor – C & Assembly: A Glimpse into the Linux Kernel

Mustafa Akgül Free Software Camp | 2025

- Delivered hands-on training in **low-level systems**, covering **computer & CPU architecture**, **C/Assembly (x86)**, **reverse engineering**, **Linux kernel internals**, and **eBPF/XDP**.

Linux Internal Structure: Introduction to BPF / XDP

Mustafa Akgül Free Software Camp | 2024

- Developed **eBPF/XDP** networking applications [↗](#), including a **Layer 4 load balancer and firewall**, with **XDP packet processing** and pinned **eBPF maps**. Modified, compiled, and tested a custom **Linux kernel (AArch64)** in **QEMU** before deployment on Raspberry Pi hardware.

Education

Bachelor's Degree

Ostim Technical University [↗](#)

2021/09 – 2025/06 | Software Engineering

GPA: 3.28

References

Python Triage Member / Software Engineer [↗](#)

Furkan Onder, Rapyuta Robotics

E-mail: furkanonder@protonmail.com [↗](#)

Real-Time Operating System Developer [↗](#)

Furkan Sonkaya, Tubitak BILGEM

E-mail: furkansonkaya61@gmail.com

Organizations

Linux Users Association [↗](#)

Association Member - Linux Enthusiast